ELIOS MERNICA

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Staff / Principal Full Stack & AI Architect with elite expertise in scaling distributed microservices, deterministic agentic workflows, and production-grade LLM systems. Proven executive capability in driving technical roadmaps, optimizing token context economics, implementing enterprise RAG/vector architectures, and competitive hackathon leadership. Expert across Python (FastAPI), Node.js, TypeScript, React, Kubernetes, and distributed event-driven data stores.

PRINCIPAL TECHNICAL EXPERIENCE & LEADERSHIP

Principal Full Stack & AI Systems Architect **2022 – Present**

Independent Principal Engineering Consultant | Remote

- Spearheaded the design and zero-downtime deployment of **15+ enterprise LLM systems** using Python (FastAPI), Node.js, and TypeScript, engineering robust rate-limiting guardrails, prompt caching layers, and end-to-end telemetry (OpenTelemetry, Prometheus).
- Architected fault-tolerant autonomous multi-step agentic execution engines utilizing **LangChain, LangGraph, and Anthropic Claude API**, integrating Model Context Protocol (MCP) servers and deterministic function-calling frameworks with sub-500ms inference latencies.
- Optimized enterprise Retrieval-Augmented Generation (**RAG**) pipelines, hybrid vector database indexes (PostgreSQL pgvector, MongoDB Atlas Vector Search), and chunking strategies, improving precision-at-k by **42%**.
- Mentored cross-functional engineering squads, established company-wide secure coding standards, automated CI/CD security scanning, and drove cloud infrastructure cost optimization (AWS/GCP), reducing operational cloud spend by **35%**.

Technical Support & Global Client Lead **2024**

Linkify

- Authored critical developer documentation, resolved complex distributed integration blockers, and managed strategic communications for international enterprise clients in native-level English.

KEY PROJECTS & HACKATHON ACHIEVEMENTS (CHRONOLOGICAL)

Genpact Hackathon 2026 (Kosovo) — 3rd Place **2026**

Architected an enterprise-grade AI healthcare diagnostic triage platform featuring streaming clinical records ingestion, FHIR compliance, and sub-second decision support.

Python, FastAPI, React, PostgreSQL, LLMs, Docker

Innovation Academy Hackathon — 1st Place (AI) & Eltrix Cloud **June 2025**

Won 1st Place in AI with Eltrix Cloud, a distributed creative content platform powered by Anthropic Claude API, LangChain & LangGraph with pgvector semantic indexing and automated RAG workflows.

FastAPI, Python, React, TypeScript, LangChain, Claude API, PostgreSQL

A++ (APL) — Custom AI Language & Transpiler **2024 – 2025**

Engineered a robust programming language compiler with an AI-assisted live transpiler, AST parser, interactive web playground, CLI toolchain, and secure containerized execution sandboxing.

Python, FastAPI, React, Node.js, WebGL

ESMP — Enterprise Marketplace OS **2024**

High-throughput e-commerce marketplace infrastructure with real-time streaming analytics, automated ledger invoicing, distributed inventory locking, and secure Stripe webhooks.

React, TypeScript, Node.js, PostgreSQL, MongoDB, Redis

Foodery — Smart Delivery Ecosystem **2024**

Scalable microservices delivery ecosystem featuring real-time GPS websocket tracking, geospatial dispatcher routing algorithms, and reactive executive dashboards.

Next.js, TypeScript, Supabase, PostgreSQL, WebSockets

ForexPulse & Calista — Enterprise Portals **2024**

Financial market telemetry engine with high-frequency WebSocket rate feeds and high-performance luxury e-commerce portals featuring GPU-accelerated 3D WebGL rendering.

JavaScript, React, Three.js, Tailwind CSS, WebSockets

TECHNICAL EXPERTISE & CORE COMPETENCIES

Languages & Core: TypeScript, JavaScript, Python, Node.js, C#, SQL, HTML5, CSS3, Bash

Distributed Systems & Architecture: Microservices, Event-Driven Architecture, REST, GraphQL, WebSockets, gRPC, Docker, Kubernetes, CI/CD

Advanced AI, LLM & Agentic Ecosystems: Anthropic API (Claude), OpenAI API, LangChain, LangGraph, RAG, Autonomous Agents, Prompt Caching, Vector Search, MCP, Fine-Tuning

Databases, Caching & Cloud Infrastructure: PostgreSQL (pgvector), MongoDB, MongoDB Atlas, Redis, Supabase, Firebase, AWS, Vercel, OpenTelemetry, Prometheus

Spoken Languages: Albanian (Native), English (Fluent / Native American Professional Proficiency)

EDUCATION & PROFESSIONAL CREDENTIALS

B.Sc. in Computer Science
Isa Boletini – UBMI (2025 – Present)

Web & Software Engineering Certification
Unity Tech Hub & Innovation Academy